Figure 1: Table Structure (`DESCRIBE purchases`)

```
mysql> DESCRIBE purchases;
+------------------+---------------+------+-----+---------+----------------+
| Field            | Type          | Null | Key | Default | Extra          |
+------------------+---------------+------+-----+---------+----------------+
| customer_id      | int           | NO   | PRI | NULL    | auto_increment |
| customer_name    | varchar(100)  | NO   |     | NULL    |                |
| purchase_date    | date          | NO   | MUL | NULL    |                |
| product_id       | varchar(20)   | NO   |     | NULL    |                |
| product_category | varchar(50)   | NO   | MUL | NULL    |                |
| amount           | decimal(10,2) | NO   |     | NULL    |                |
+------------------+---------------+------+-----+---------+----------------+
6 rows in set (0.117 sec)
```

Figure 2: Database Indexes (`SHOW INDEX FROM purchases`)

```
mysql> SHOW INDEX FROM purchases;
+-----------+------------+----------------------+--------------+------------------+-----------+-------------+----------+--------+------+------------+---------+------------
--+---------+------------+
| Table     | Non_unique | Key_name             | Seq_in_index | Column_name      | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_commen
t | Visible | Expression |
+-----------+------------+----------------------+--------------+------------------+-----------+-------------+----------+--------+------+------------+---------+------------
--+---------+------------+
| purchases |          0 | PRIMARY              |            1 | customer_id      | A         |          30 |     NULL |   NULL |      | BTREE      |         |
  | YES     | NULL       |
| purchases |          1 | idx_product_category |            1 | product_category | A         |           4 |     NULL |   NULL |      | BTREE      |         |
  | YES     | NULL       |
| purchases |          1 | idx_purchase_date    |            1 | purchase_date    | A         |           5 |     NULL |   NULL |      | BTREE      |         |
  | YES     | NULL       |
+-----------+------------+----------------------+--------------+------------------+-----------+-------------+----------+--------+------+------------+---------+------------
--+---------+------------+
3 rows in set (3.081 sec)
```

Figure 3: Total Records (`SELECT COUNT(*)`)

```
mysql> SELECT COUNT(*);
+----------+
| COUNT(*) |
+----------+
|        1 |
+----------+
1 row in set (0.023 sec)

mysql>
```

Figure 4: Date Range Query Result

```
mysql> SELECT *
    -> FROM purchases
    -> WHERE purchase_date BETWEEN '2026-06-01' AND '2026-06-03';
+-------------+---------------+---------------+------------+----------------------+-----------+
| customer_id | customer_name | purchase_date | product_id | product_category     | amount    |
+-------------+---------------+---------------+------------+----------------------+-----------+
|           1 | John Okafor   | 2026-06-01    | PRD1001    | Electronics          | 125000.00 |
|           2 | Mary Eze      | 2026-06-01    | PRD1002    | Food                 |   8500.00 |
|           3 | David Bello   | 2026-06-01    | PRD1003    | Clothing and Textile |  22000.00 |
|           4 | Grace Ahmed   | 2026-06-02    | PRD1004    | Health and Beauty    |  15500.00 |
|           5 | Samuel Obi    | 2026-06-03    | PRD1005    | Electronics          |  89000.00 |
+-------------+---------------+---------------+------------+----------------------+-----------+
5 rows in set (0.038 sec)

mysql>
```

Figure 5: Product Category Query Result

```
mysql> SELECT *
    -> FROM purchases
    -> WHERE product_category = 'Electronics';
+-------------+---------------+---------------+------------+------------------+-----------+
| customer_id | customer_name | purchase_date | product_id | product_category | amount    |
+-------------+---------------+---------------+------------+------------------+-----------+
|           1 | John Okafor   | 2026-06-01    | PRD1001    | Electronics      | 125000.00 |
|           5 | Samuel Obi    | 2026-06-03    | PRD1005    | Electronics      |  89000.00 |
|          13 | Peter Chukwu  | 2026-07-08    | PRD1013    | Electronics      |  56000.00 |
|          17 | Victor Akin   | 2026-07-08    | PRD1017    | Electronics      |  76000.00 |
|          21 | Kelvin Abiola | 2026-07-08    | PRD1021    | Electronics      | 149000.00 |
|          25 | Henry Ogbonna | 2026-07-08    | PRD1025    | Electronics      | 285000.00 |
|          29 | Andrew Ekanem | 2026-07-09    | PRD1029    | Electronics      |  99000.00 |
+-------------+---------------+---------------+------------+------------------+-----------+
7 rows in set (0.023 sec)

mysql>
```

Figure 6: Python Search Simulation Output

```
C:\Users\KDNS\Documents\MIVA\300 Level\2nd Semester\Data Management\Mid-Semester Assessment>python retail_business.py
Total Records Loaded: 30

==============================
Sequential Search
==============================
{'Timestamp': '6/30/2026 18:01:32', 'Customer Name': 'Samuel Obi', 'Purchase Date  ': '6/3/2026', 'Product ID': 'PRD1005', 'Product Category': 'Electronics', 'Amount': '890
00'}
Execution Time: 0.0000075000 seconds

==============================
Binary Search
==============================
{'Timestamp': '6/30/2026 18:01:32', 'Customer Name': 'Samuel Obi', 'Purchase Date  ': '6/3/2026', 'Product ID': 'PRD1005', 'Product Category': 'Electronics', 'Amount': '890
00'}
Execution Time: 0.0000088000 seconds

==============================
Performance Comparison
==============================
Sequential Search is Faster

C:\Users\KDNS\Documents\MIVA\300 Level\2nd Semester\Data Management\Mid-Semester Assessment>
```